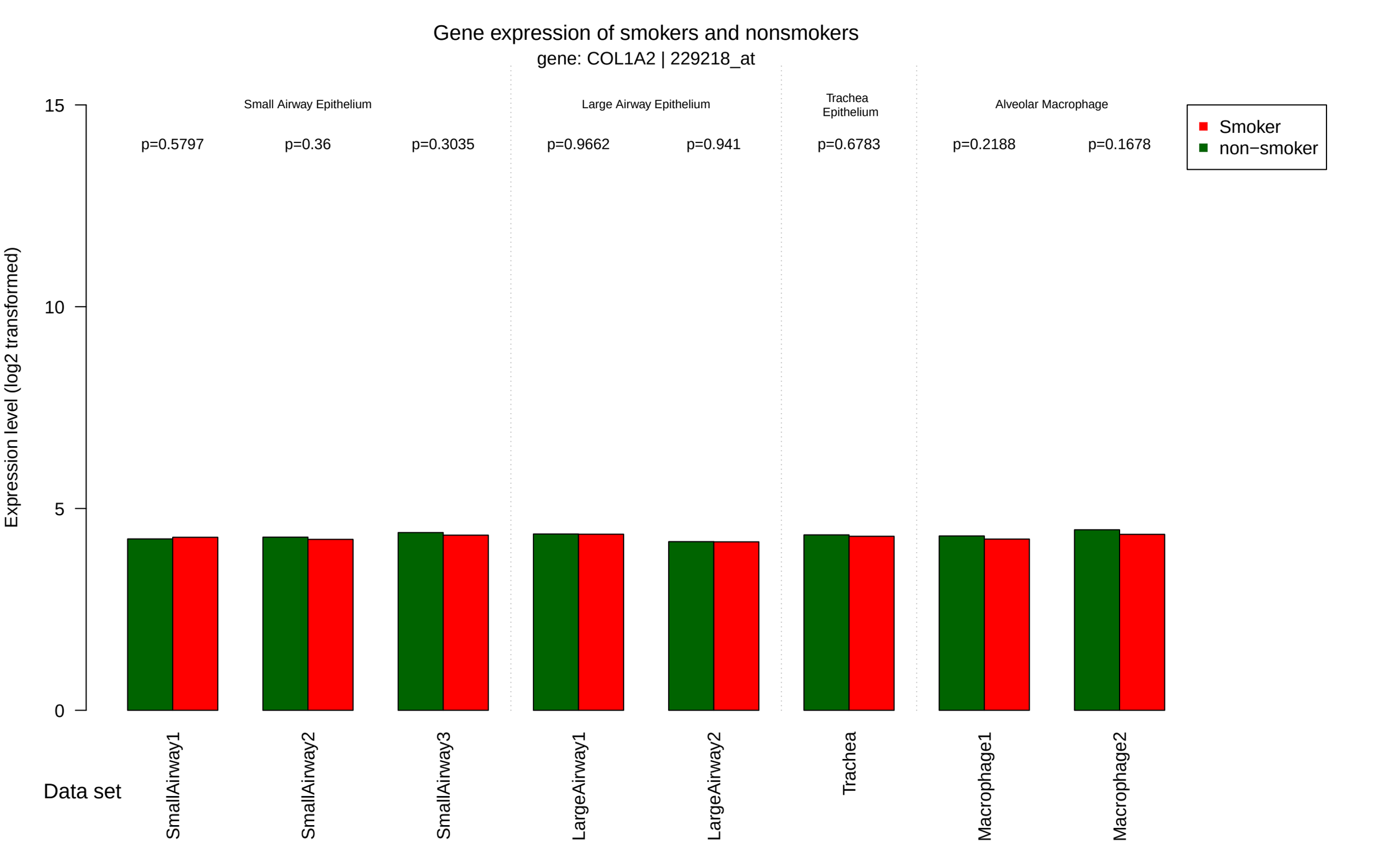
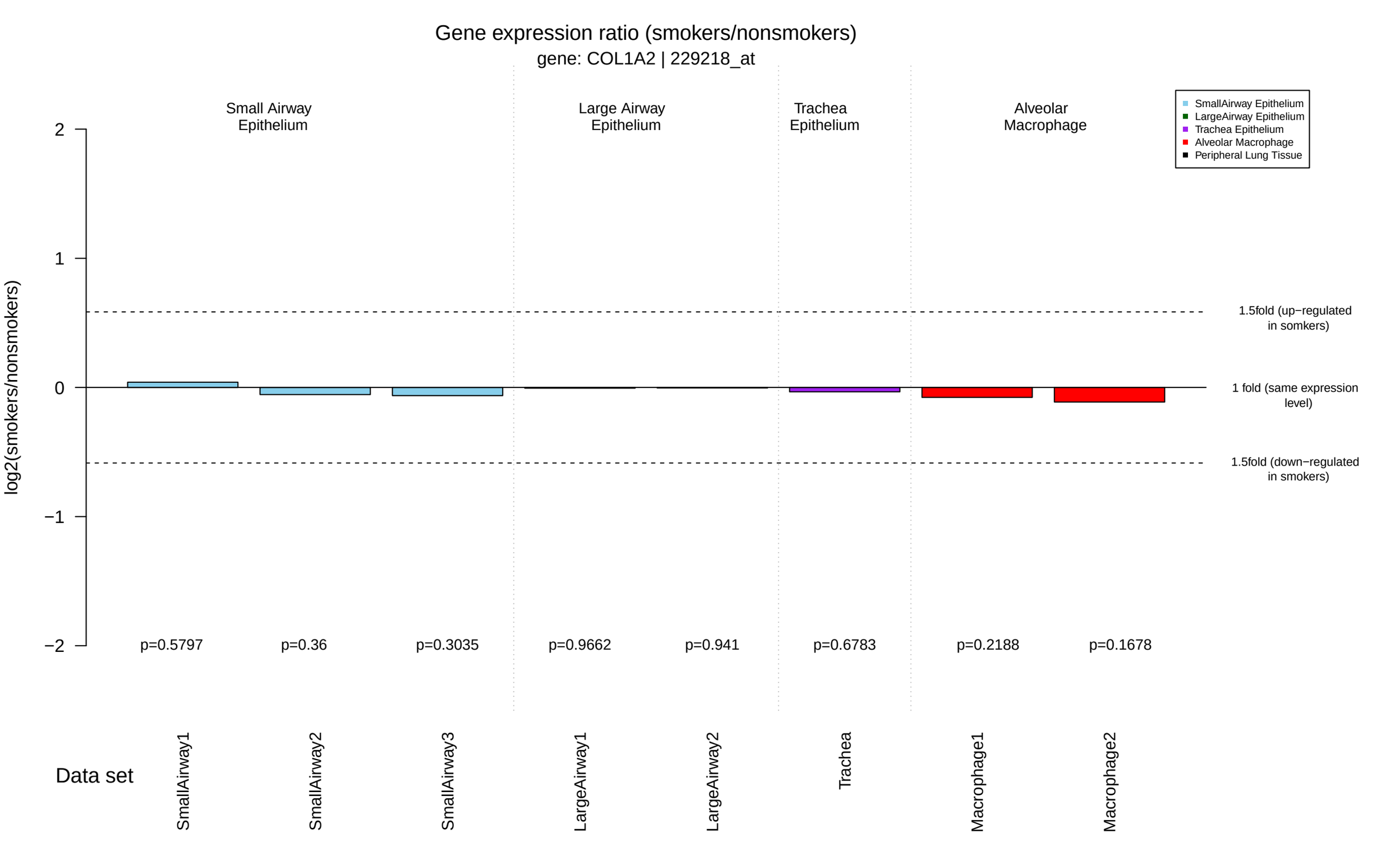
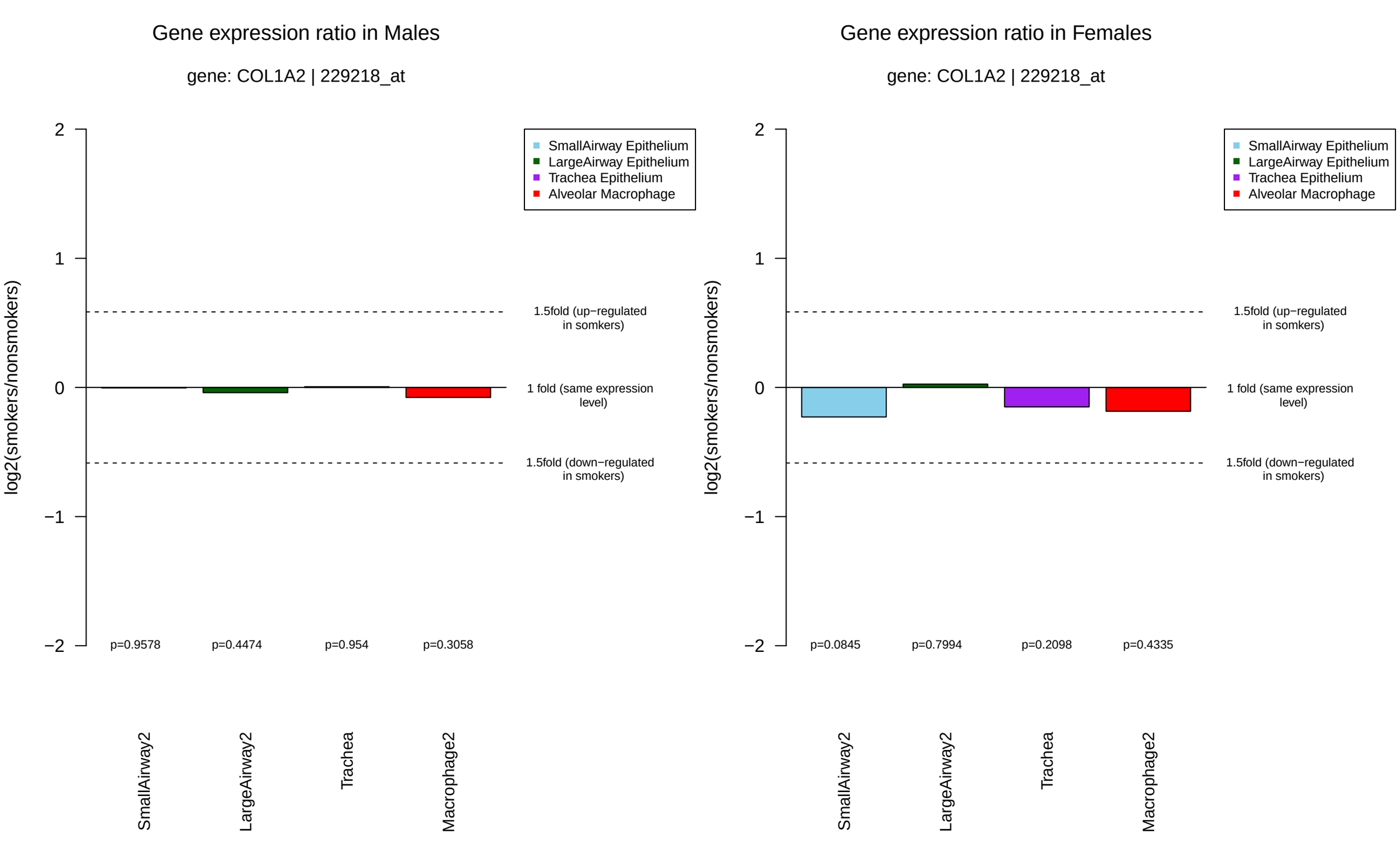
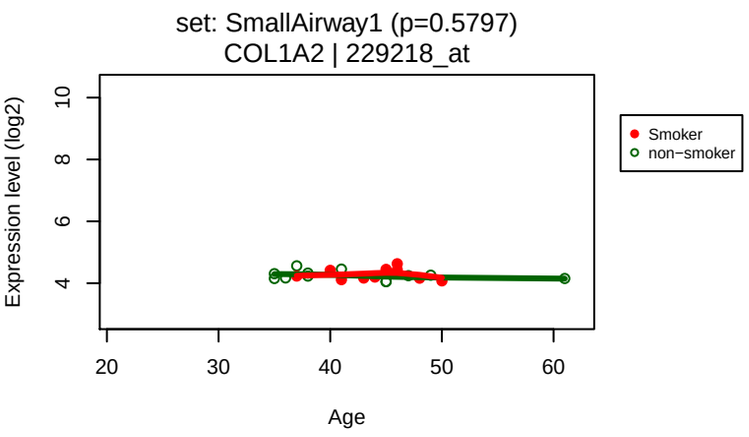
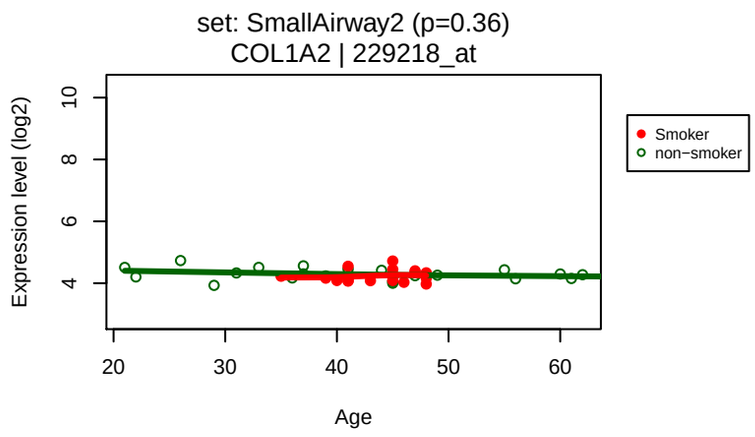
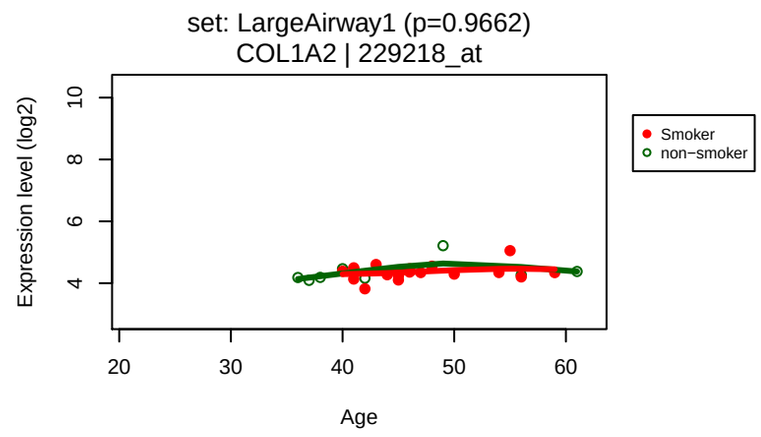

Gene expression of smokers and nonsmokers
gene: COL1A2 | 229218_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5797
p=0.36
p=0.3035
p=0.9662
p=0.941
p=0.6783
p=0.2188
p=0.1678
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

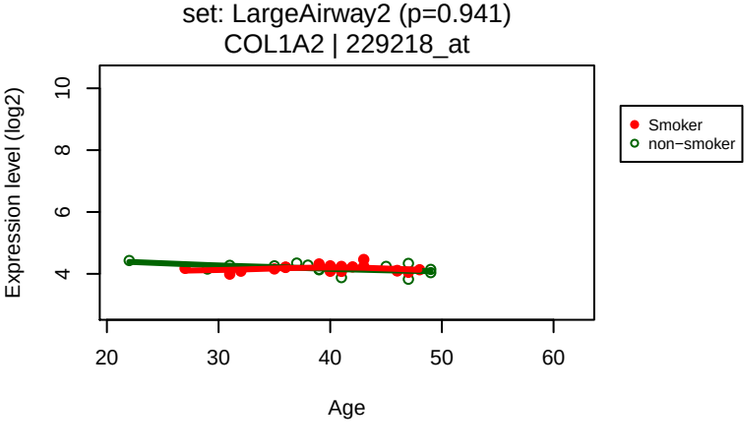

Gene expression ratio (smokers/nonsmokers)
gene: COL1A2 | 229218_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5797
p=0.36
p=0.3035
p=0.9662
p=0.941
p=0.6783
p=0.2188
p=0.1678
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

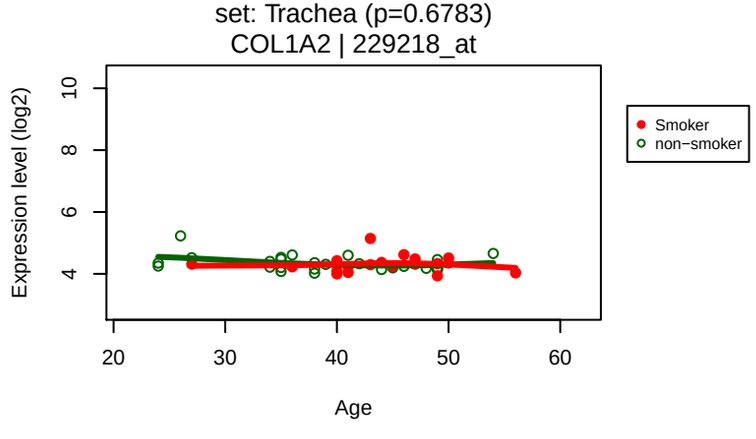

Gene expression ratio in Males
gene: COL1A2 | 229218_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9578
p=0.4474
p=0.954
p=0.3058
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL1A2 | 229218_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0845
p=0.7994
p=0.2098
p=0.4335
SmallAirway2
LargeAirway2
Trachea
Macrophage2

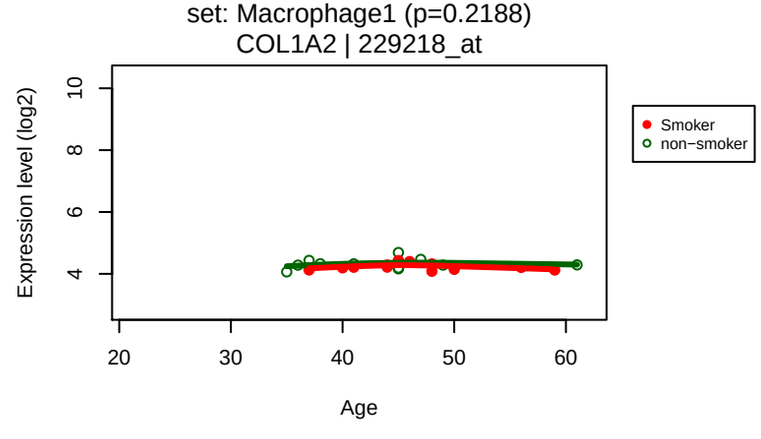

set: SmallAirway1 (p=0.5797)
COL1A2 | 229218_at
Expression level (log2)
Age
Smoker
non−smoker

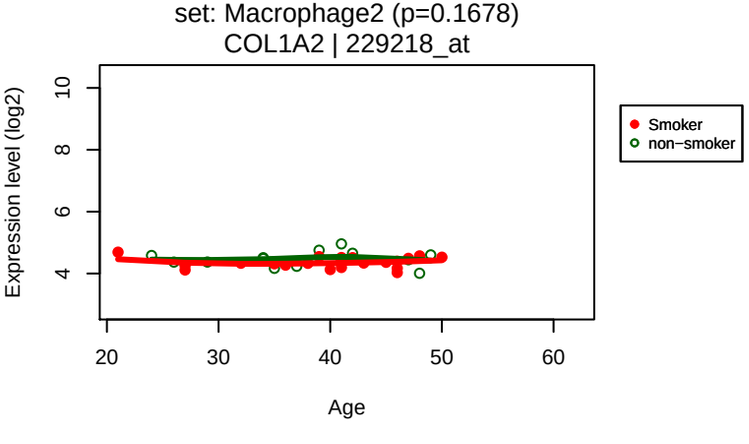

set: SmallAirway2 (p=0.36)
COL1A2 | 229218_at
Expression level (log2)
Age
Smoker
non−smoker

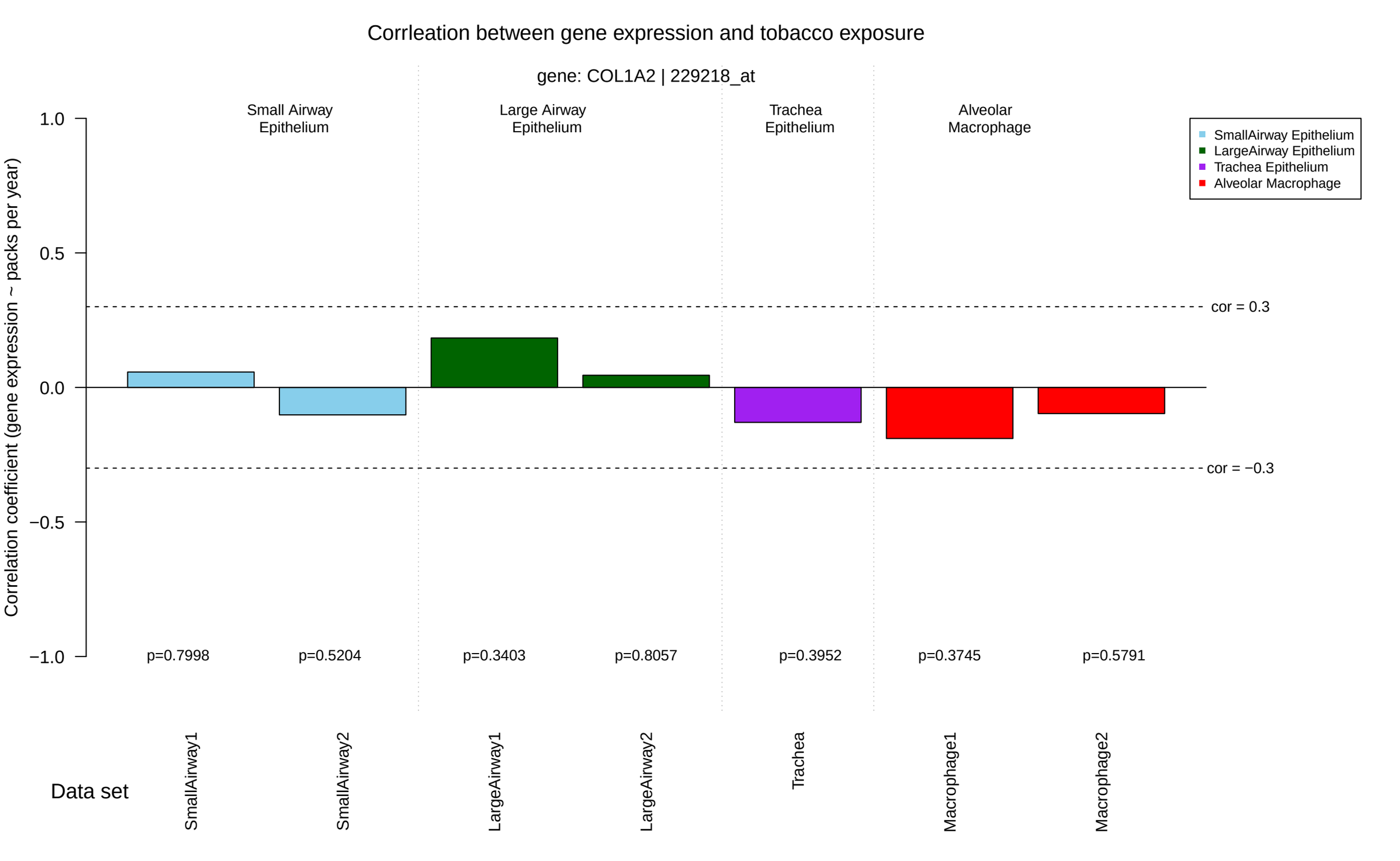

set: LargeAirway1 (p=0.9662)
COL1A2 | 229218_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.941)
COL1A2 | 229218_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6783)
COL1A2 | 229218_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2188)
COL1A2 | 229218_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1678)
COL1A2 | 229218_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL1A2 | 229218_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7998
p=0.5204
p=0.3403
p=0.8057
p=0.3952
p=0.3745
p=0.5791
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2